Examiner only

**3.** The halogens are a group of elements found in group 7 of the periodic table. The table below gives some information about the elements in this group.

| Halogen | Atomic number | Melting point (°C) | Boiling point (°C) | Radius of the atom (units) |
|---|---|---|---|---|
| fluorine | 9 | –219 | –188 | (0.64 / 1.1 / 1.7) |
| chlorine | 17 | **(–5 / –101 / –220)** | –34 | 0.99 |
| bromine | 35 | –7 | 59 | **(1.14 / 1.45 / 1.56)** |
| iodine | **(22 / 31 / 53)** | 114 | 184 | 1.33 |
| astatine | 85 | 302 | 380 | 1.40 |

*(a)* **Complete** the table above by **circling** the correct value in each of the brackets. One has been done as an example. [3]

*(b)* Use the information in the table and your own knowledge to answer the following questions.

(i) The boiling points of the alkali metals (group 1) decreases from the top to the bottom of their group. Describe whether the halogens follow the same trend. [2]

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(ii) State the **two** halogens in the table which are gases at 20°C. [2]

.................................................. and ..................................................

(iii) State which halogen is a liquid at 20°C. [1]

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(iv) State what happens to the size of the halogen atoms as you go down from the top to the bottom of the group. [1]

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(v) **Tick** (✓) the box next to the correct statements about halogens. [3]

- Fluorine has the highest melting point ☐
- The most reactive halogen is fluorine ☐
- Astatine has the most electrons ☐
- Iodine is a solid at 20°C ☐
- The most reactive halogen is astatine ☐
- Chlorine boils at a lower temperature than fluorine ☐

*(c)* Sodium chloride can be made by burning sodium in chlorine gas. However it is found naturally in rock salt, which can be obtained from the Earth's crust. **Tick** (✓) the box next to the correct process for extraction of rock salt from the Earth's crust. [1]

- fracking ☐
- deep shaft mining ☐
- fractional distillation ☐
- drilling ☐

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